Bank Marketing Data Analysis

Telemarketing campaign for term deposits · 11,162 customers · Beginner-friendly report

1. Dataset Overview

Each row is one customer who was called by the bank. The goal: did the customer subscribe to a **term deposit** (`deposit` = yes/no)?

11,162	17	5,289	47.4%
Customers (rows)	Columns	Subscribed	Overall success rate

Group	Columns
Customer profile	age, job, marital, education
Financial	default (credit default), balance (yearly average, €), housing (home loan), loan (personal loan)
Current campaign	contact, day, month, duration (call seconds), campaign (calls this campaign)
Past campaign	pdays (days since last contact), previous (earlier contacts), poutcome (earlier result)
Target	deposit (yes / no)

Average age 41 · median balance €550 · average call 372 seconds (about 6 minutes) · average 2.5 calls per customer.

2. Data Cleaning

Check	Result	Action
Duplicate rows	0	None needed
Data types	Numbers and text are correct	None needed
Negative balances	688 customers (overdrawn accounts)	Kept: valid values
Very high balances	Top 1% above €13,227 (max €81,204)	Kept, but use the median, not the average
Ages	18 to 95; 55 customers over 80	Kept: realistic
pdays = -1	8,324 customers never contacted before	Kept: -1 means "not contacted"
Text value "unknown"	See next section	Kept as its own category

3. Missing Value Analysis

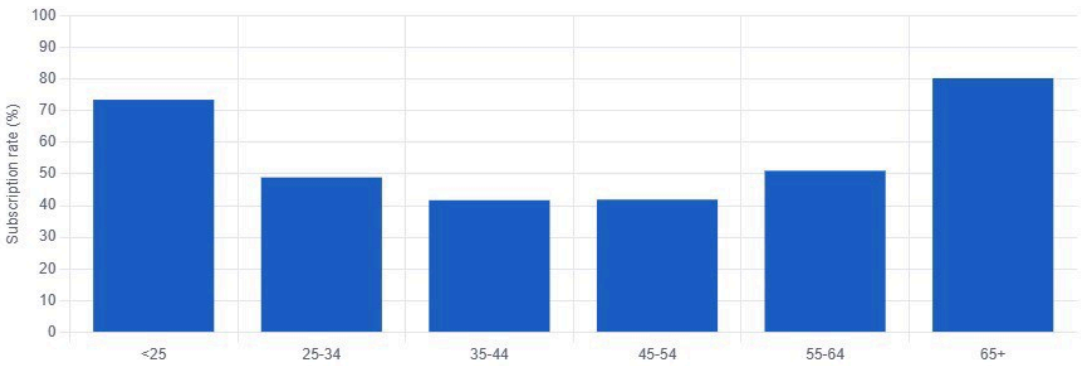
There are **no blank (null) cells** in any column. However, some columns use the word **"unknown"** as a hidden missing value:

Column	"unknown" count	Share
poutcome (previous outcome)	8,326	74.6%
contact	2,346	21.0%
education	497	4.5%
job	70	0.6%

> Most "unknown" poutcome values are customers never contacted before, so this is normal. Contact type "unknown" has a very low success rate (22.6%), which is worth noting.

4. Customer Age Analysis

Age group	Customers	Subscription rate
Under 25	282	73.4%
25–34	3,628	48.9%
35–44	3,366	41.7%
45–54	2,205	41.9%
55–64	1,256	51.0%
65+	425	80.2%

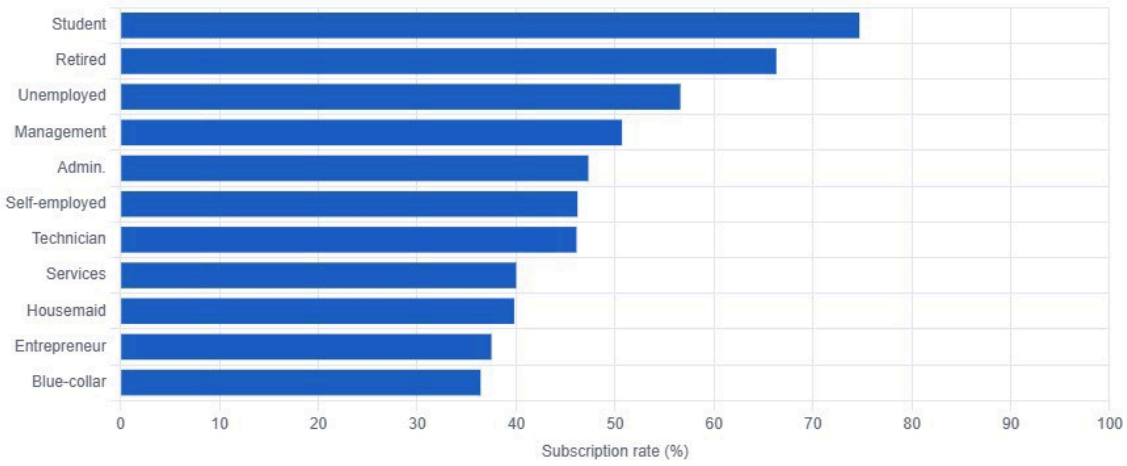


Most customers are 25–44, but they are **not** the most likely to subscribe. Young (under 25) and senior (65+)

5. Job-wise Customer Analysis

Job	Customers	Subscription rate
Student	360	74.7%
Retired	778	66.3%
Unemployed	357	56.6%
Management	2,566	50.7%
Admin.	1,334	47.3%
Self-employed	405	46.2%
Technician	1,823	46.1%
Services	923	40.0%
Housemaid	274	39.8%
Entrepreneur	328	37.5%
Blue-collar	1,944	36.4%

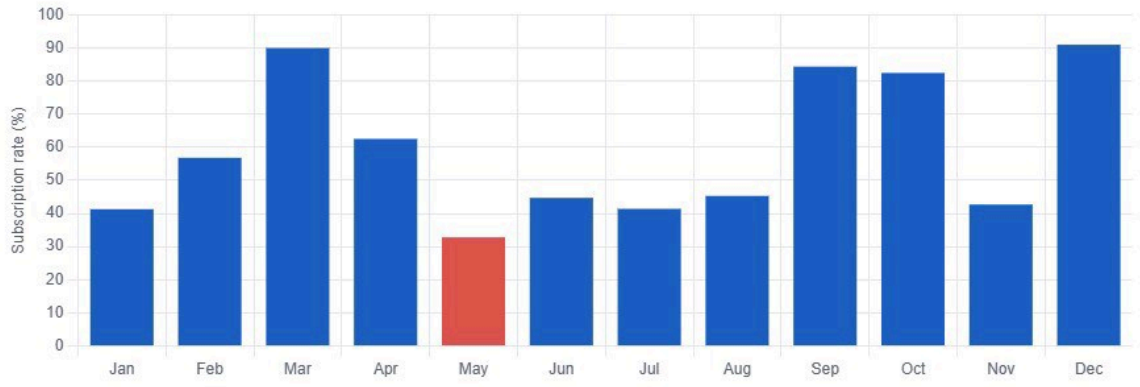
Unknown	70	48.6%



Management is the biggest group (23% of customers) and converts above average. Blue-collar is the second biggest group but converts the least.

6. Campaign Success Rate

By month (the bank called most in May, but May performed worst):



By number of calls to the same customer:

Calls	Customers	Success rate
1	4,798	53.4%
2	3,028	46.3%
3	1,321	46.8%
4–5	1,149	39.7%
6–10	656	31.4%
11+	210	22.4%

By contact method and previous campaign result:

Group	Customers	Success rate
Cellular	8,042	54.3%
Telephone	774	50.4%
Contact unknown	2,346	22.6%
Previous campaign: success	1,071	91.3%
Previous campaign: other	537	57.2%
Previous campaign: failure	1,228	50.3%
Previous campaign: unknown (new)	8,326	40.7%

> Calling again and again does not help: after 5 calls, success falls sharply. Past success is the strongest signal: 9 out of 10 earlier successes subscribed again.

7. Subscription Analysis

47.4%
Said Yes (5,289)

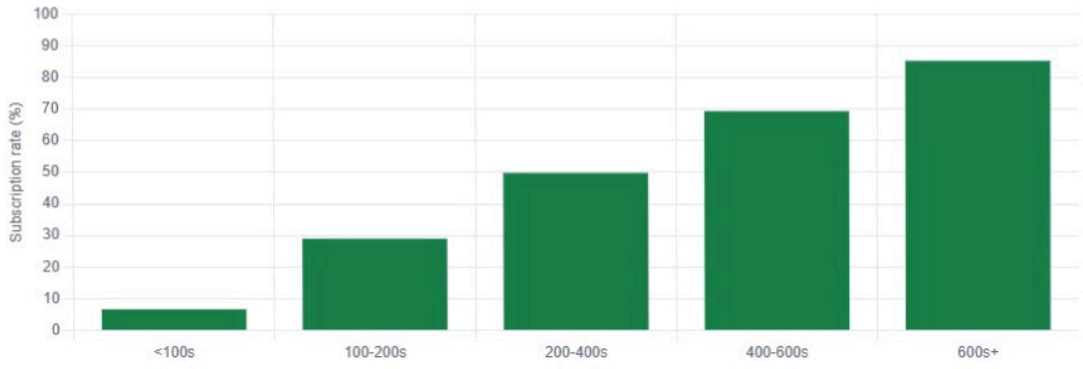
52.6%
Said No (5,873)

The data is well balanced between Yes and No, which makes it good for analysis and prediction.

Customer type	Yes rate
No home loan / has home loan	57.0% / 36.6%
No personal loan / has personal loan	49.5% / 33.2%
No credit default / has default	47.6% / 31.0%
Single / Divorced / Married	54.3% / 48.1% / 43.4%

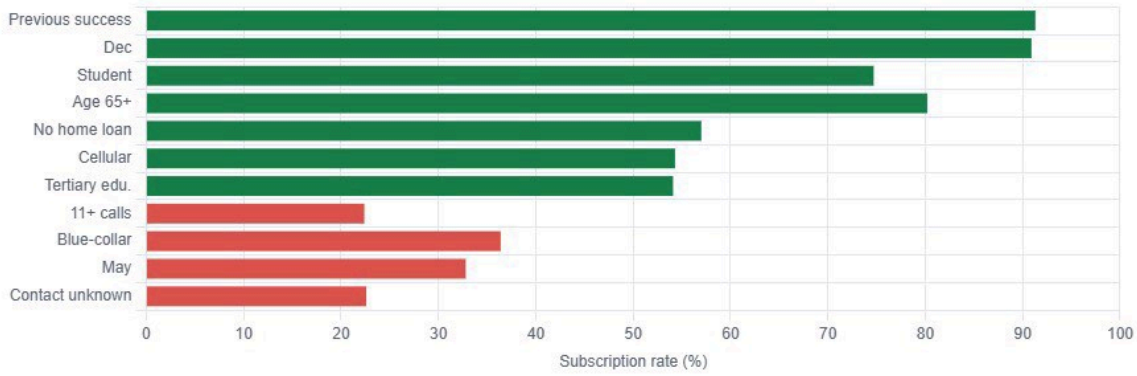
Average	Subscribed	Did not subscribe
Account balance	€1,804	€1,280
Call length	537 sec	223 sec
Calls in campaign	2.1	2.8

Success rate by call length:



8. Data Visualization

The charts are placed in sections 4, 5, 6 and 7. Here is one more overview of the most influential factors, comparing the highest and lowest success rate in each factor:



9. Key Business Insights

- **Half of the calls succeed.** The overall rate is 47.4%, a healthy baseline.
- **Past success predicts future success.** 91.3% of customers who said yes earlier subscribed again.
- **Students, retirees, and the 65+ group are the best segments** (66–80% success).
- **May is overused.** 25% of all calls were in May with only 32.8% success. March, September, October and December exceed 82%, but received only 1,100 calls (about 10%).
- **More calls means less success.** Customers called more than 5 times subscribe at 22–31%.
- **Contact type "unknown" performs poorly (22.6%)**, and affects 21% of customers.
- **Loans reduce interest.** Customers with home or personal loans subscribe 16–20 points less.
- **Engaged customers say yes.** Calls over 10 minutes reach 85% success; calls under 100 seconds only 7%.

10. Business Recommendations

#	Recommendation	Why
1	Re-contact customers who subscribed in a past campaign first	91% success
2	Target students, retirees and seniors with tailored offers	66–80% success
3	Cap attempts at 3–4 calls per customer	Success drops below 40% a
4	Spread calls across the year; test more campaigns in March, September, October and December	May is crowded and weakes
5	Prefer cellular calls and record the contact type properly	54% vs 23% for unknown
6	Give lower priority to customers with several loans or defaults, or use a different pitch	Lower interest
7	Train agents to keep customers engaged in the first minutes of the call	Longer calls are far more su
8	Build a simple scoring model (age, job, past result, loans) to rank who to call	Saves cost and time

Note: these are patterns in the data, not proof of cause. Test changes on a small group before a full rollout.